

ITI5202 – DATA PROCESSING FOR BIG DATA

ASSESSMENT 2 PHASE 1 – PROJECT PROPOSAL

Project Title:

**Predicting Used Cars Price in the US
based on Its Condition and Features**

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1. Project Description

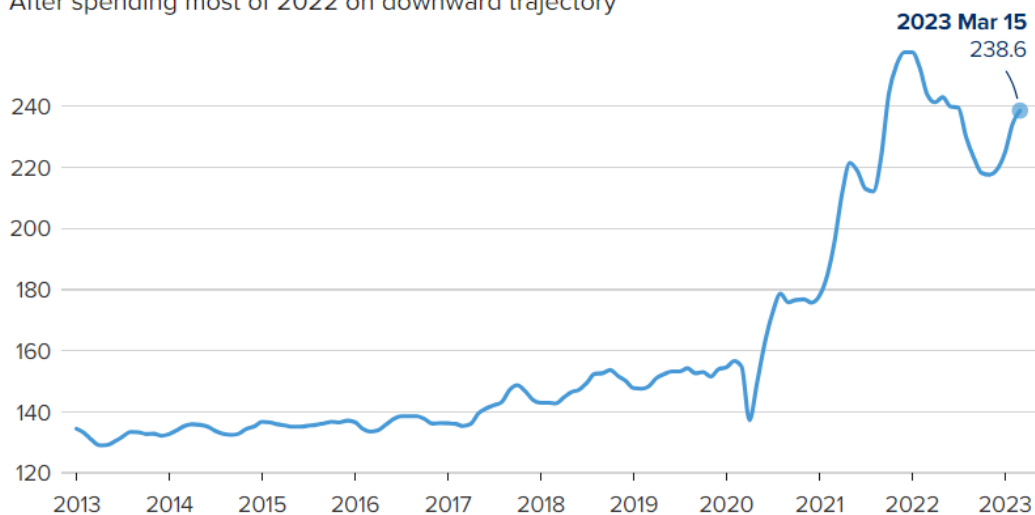
1.1. Background

Many people may think that owning a car is just a secondary or even tertiary need. On the contrary, in the US, it is crucial to own a car because the community and infrastructure depend on cars as the primary means of traveling or commuting. This statement is based on an article written by Inkstick Media which states that “Car ownership has become a de facto prerequisite for Americans to access the labor market and engage in civic life” ^[1]. Nonetheless, not everyone can buy a car as car prices can be relatively expensive. That is why, some people may prefer to buy used cars instead of new cars, which are cheaper compared to brand-new cars.

Sadly, due to the adverse effect of the COVID-19 pandemic on the economic sector, used car prices skyrocketed in recent years (can be seen in Figure 1). The main cause for this problem is that the production of new cars has mostly ceased due to the lack of semiconductor chips available ^[2]. This creates a problem for people who still haven’t owned a car yet or their car is broken as it becomes harder for them to acquire a new car. A woman in an interview with the New York Times, Ms. Cote, 29, said “Food’s gone up a bit, but the cost of housing, health care, and cars is pretty brutal.” Because of the rise in car prices, she became unable to make any savings because all of her income was spent on her necessities ^[3].

Used car prices are rising again

After spending most of 2022 on downward trajectory



Note: January 1997 = 100

Chart: Gabriel Cortes / CNBC

Source: [Manheim Used Vehicle Value Index](#)



Figure 1. Used cars price trend in the US ^[2]

1.2. Problem Formulation

Because the rise in used car prices was caused by the COVID-19 pandemic, it is only natural to expect that their price will be back to normal after the pandemic passes. Nevertheless, in an interview with CNBC, Charlie Chesbrough, senior economist and senior director of Industry Insights at Cox Automotive said “There is still a lot of demand out there that just has not been met yet. It’s unsatiated demand. Until the new market can rebuild such that demand is met and that there is enough product out there for everybody, the U.S. markets are going to continue to be doing very robust sales”. Without new vehicles, there are no used vehicles. Therefore, in the worst-case scenario, it is even possible that the prices may rise again by 60% like when COVID-19 happened ^[4].

To make matters worse, nowadays people prefer to buy cars online without even going to the dealer and inspect the car themselves before buying it beforehand. Based on the data from AA Cars, one-in-five (21%) people bought a car without seeing it first in 2022 compared to 2021, which was only just one-in-ten (10%) ^[5]. This phenomenon gives rise to a problem as the cars bought might not be the same as what they had in mind, thus it is more likely for the dealer to outsmart the buyer this way. Although this can be mitigated by reading online articles that advise about how to buy used cars, it would be better if there is a way to know the correct prices of the car reliably. Thus, it may help the buyer to be able to make good purchases. One way of doing this is by creating a model to predict the used car prices based on the car’s condition using machine learning such as linear regression.

1.3. Past Research

Thakkar et al. (2023) have done research about predicting car prices using linear regression, which is considered to be a popular way to do it ^[6]. From their research, they found that linear regression is indeed suitable for predicting car prices with an accuracy ranging from 78% - 86.5%. In addition, they also stated that typically, the models are made by using the car’s manufacture, kms_driven, and condition as the predictors, and some researchers even do feature engineering on the predictors to improve the prediction. Overall, they said that “the model architecture for old car price prediction using linear regression is relatively straightforward, with the primary focus on selecting appropriate input variables and optimizing the coefficients through the training process” ^[6].

1.4. Objective

The main purpose of this project is to do a replication study on the method done by Thakkar et al. (2023) on a different data set and possibly also on a different set of predictors. The new dataset we’re going to use is from the used car data set in the US that has 3 million records which makes it larger than what Thakkar et al. have used, which was around 5,500 records. Moreover, other than creating a linear regression model, we also plan to do a feature analysis on the resulting model to determine which factors have a greater impact on the car’s price.

2. Business Model

2.1. Business Area and Benefits

The sector that will benefit the most from this research is the automotive industry, especially the used car market. Moreover, the benefit will not only be limited to just the dealers but also the potential buyers and even the car owners who want to sell their cars. The resulting linear regression model can be used to determine the price of the car fairly based on its features and condition. This price then can be used as a benchmark for the seller to determine the initial price or even the buyers can use this price to negotiate if the initial price is too expensive. In addition, the feature analysis done will help to determine which factor has a great impact on the price of the car by seeing its variable importance. With this knowledge, car owners who potentially sell their cars in the future can know which feature or condition of the car they need to keep well so that the price stays high.

2.2. Stakeholders

- **Used car dealers:**
Dealers can use the resulting model to determine the price of the car that they are going to sell. By using the calculated price as a benchmark, the dealer can safely decide on the price without “damaging” the used car market by setting the price too low or too high.
- **Car consultants:**
Car/automobile consultants are people commissioned by either dealers or buyers to provide information necessary for them to deal with the cars they are interested in. Based on the feature analysis result, the consultant can also use it to guide the buyers about what features or specifications are important.
- **Potential buyers:**
The potential buyers can easily check whether the price given on the car that they want to buy is following the market’s price or not by the car’s features and conditions. This is done to avoid potential scamming from the seller by setting the price too high.
- **Car owners who want to sell their car:**
For car owners, the usage of the model is a combination between dealers and buyers. They can use the model to set the price appropriately and if they sell it to a third party (dealer), they can know whether the price set by them is suitable or not.

2.3. Challenges

Challenges	Mitigation
The raw dataset's size is too big (± 3 million rows). This may result in a slower processing speed due to the large data.	Apply parallel processing by partitioning the data.
Some of the fields (columns) do not have a proper data dictionary.	Discard the columns that are hard to guess. Use assumptions on the clear ones.
The dataset has a large number of missing values. However, imputing specific values on some columns is not possible as each car has its own conditions or features.	If the data type is Boolean (True/False), we can just impute False as the default value. On the other hand, for other columns, we can just drop the missing values as the remaining records should be enough for training a model considering the size of the dataset.

3. Data and Data Analysis

3.1. Data Source and Characteristic

The dataset we used was obtained from Kaggle (kaggle.com/datasets/ananyamital/us-used-cars-dataset/data) and was downloaded on October 10, 2023. This dataset contains around **3 million** used car sales records from the US. The author of the dataset, Ananay Mital, acknowledged that they obtained the data with a self-made crawler on Cargurus inventory in September 2020. The dataset consists of 3 million rows and 66 columns. The data dictionary is provided in Table 1. However, because we decided to not use all columns, we will only list the metadata of those selected columns.

Table 1. Data dictionary of the used cars dataset

Column Name	Type	Description
vin	String	Vehicle Identification Number is a unique encoded string for every vehicle.
back_legroom	String	Legroom in the rear seat.
body_type	String	Body Type of the vehicle. Like Convertible, Hatchback, Sedan, etc.
city_fuel_economy	Float	Fuel economy in city traffic in km per litre.
daysonmarket	Integer	Days since the vehicle was first listed on the website.
engine_cylinders	String	The engine configuration. Eg: I4, V6, etc.
engine_displacement	Float	The measure of the cylinder volume swept by all of the pistons of a piston engine, excluding the combustion chambers.
franchise_make	String	The company that owns the franchise.
front_legroom	String	The legroom is in inches for the passenger seat.
fuel_tank_volume	String	Fuel tank's filling capacity in gallons.
fuel_type	String	Dominant type of fuel ingested by the vehicle.
has_accidents	Boolean	Whether the vin has any accidents registered.
height	String	Height of the vehicle in inches.
highway_fuel_economy	Float	Fuel economy in highway traffic in km per litre.
horsepower	Float	Horsepower is the power produced by an engine.
isCab	Boolean	Whether the vehicle was previously taxi/cab.
is_cpo	Boolean	Pre-owned cars certified by the dealer. Certified vehicles come with a manufacturer warranty for free repairs for a certain time period.
is_oemcpo	Boolean	Pre-owned cars certified by the manufacturer.
length	String	Length of the vehicle in inches
listing_color	String	Dominant color group from the exterior color.
mileage	Integer	-
owner_count	Integer	-

Column Name	Type	Description
power	String	-
price	Float	-
torque	String	-
transmission	String	-
wheel_system	String	-
wheelbase	String	-
width	String	-
year	Integer	-

3.2. Data Wrangling

Before we can use the dataset for analysis purposes, we need to clean the data first. First, to lessen the processing burden, we can create a subset of the data by only taking the necessary columns. Henceforward, the wrangling process will only be done on the subset.

As mentioned in the previous section, the dataset contains many missing values that need to be dealt with before using it for training a model. In our case, if the column type is Boolean/logical (True/False), such as `is_new`, `is_cpo`, `is_oemcpo`, `has_accidents`, and `isCab`, we decided to replace the NAs with a "False" value. The reason is that, for the `has_accidents` and `isCab` columns, naturally the default value is False, whereas for the other three, we assume that missing values happen because the dealer deliberately did not list those details to hide the actual age of the car or that the car did not have certification. On the other hand, for the other columns, we decided to just remove the rows that contain NAs in those columns. The reasoning behind this is that the car itself should have its own unique feature or condition, thus imputing the median or mode may distort the truth about the data. It is possible to do in-depth investigating to predict the actual value based on the other known columns. However, for this initial stage, we decided that it is not necessary as the volume of the data after removal is still big enough for training a machine learning model. The comparison of the size between the original dataset and the subset (after dealing with missing values) is provided in Figure 2.

```
> dim(cars)
[1] 3000040      66
> dim(wrangled_cars)
[1] 704408      32
```

Figure 2. Comparison of the size of the original dataset (above) and the subset (below)

Note: due to some difficulties when using PySpark in the Jupyter Notebook, the above wrangling process was done using R in RStudio. The "semi-cleaned" dataset was then written to a new CSV file that will be further processed using PySpark in Jupyter Notebook.

After dealing with missing data, the next step is to inspect the structure of the resulting data frame. We noticed that some of the columns have incorrect data types, which are listed as a string but should be numeric (integer or double). This happens because those columns contain not only the values but also their unit of measurement, such as “in” or “gal”, which are non-number characters. To convert those columns to numeric type (here we choose to convert them to double), we need to remove the non-number characters first. Then, after the values only contain numbers, we can safely convert the column type into a double type. It turns out that the dataset still contains some missing values (listed as “--“, not empty). Thus, after converting the columns to numeric, those values will turn to NA. Therefore, those records need to be removed. The comparison between the initial schema and the corrected one is provided in Figure 3.

(704408, 32)	(694605, 32)
root	root
-- vin: string (nullable = true)	-- vin: string (nullable = true)
-- back_legroom: string (nullable = true)	-- back_legroom: double (nullable = true)
-- body_type: string (nullable = true)	-- body_type: string (nullable = true)
-- city_fuel_economy: integer (nullable = true)	-- city_fuel_economy: integer (nullable = true)
-- daysonmarket: integer (nullable = true)	-- daysonmarket: integer (nullable = true)
-- engine_cylinders: string (nullable = true)	-- engine_cylinders: string (nullable = true)
-- engine_displacement: integer (nullable = true)	-- engine_displacement: integer (nullable = true)
-- franchise_make: string (nullable = true)	-- franchise_make: string (nullable = true)
-- front_legroom: string (nullable = true)	-- front_legroom: double (nullable = true)
-- fuel_tank_volume: string (nullable = true)	-- fuel_tank_volume: double (nullable = true)
-- fuel_type: string (nullable = true)	-- fuel_type: string (nullable = true)
-- has_accidents: boolean (nullable = true)	-- has_accidents: boolean (nullable = true)
-- height: string (nullable = true)	-- height: double (nullable = true)
-- highway_fuel_economy: integer (nullable = true)	-- highway_fuel_economy: integer (nullable = true)
-- horsepower: integer (nullable = true)	-- horsepower: integer (nullable = true)
-- price: double (nullable = true)	-- price: double (nullable = true)
-- width: string (nullable = true)	-- width: double (nullable = true)
-- year: integer (nullable = true)	-- year: integer (nullable = true)
-- is_cpo: boolean (nullable = true)	-- is_cpo: boolean (nullable = true)
-- is_oemcpo: boolean (nullable = true)	-- is_oemcpo: boolean (nullable = true)
-- is_new: boolean (nullable = true)	-- is_new: boolean (nullable = true)
-- listing_color: string (nullable = true)	-- listing_color: string (nullable = true)
-- mileage: integer (nullable = true)	-- mileage: integer (nullable = true)
-- owner_count: integer (nullable = true)	-- owner_count: integer (nullable = true)
-- power: string (nullable = true)	-- power: string (nullable = true)
-- torque: string (nullable = true)	-- torque: string (nullable = true)
-- isCab: boolean (nullable = true)	-- isCab: boolean (nullable = true)
-- transmission: string (nullable = true)	-- transmission: string (nullable = true)
-- wheel_system: string (nullable = true)	-- wheel_system: string (nullable = true)
-- length: string (nullable = true)	-- length: double (nullable = true)
-- wheelbase: string (nullable = true)	-- wheelbase: double (nullable = true)
-- model_name: string (nullable = true)	-- model_name: string (nullable = true)

Figure 3. The comparison of the original schema and the corrected one

3.3. Exploratory Data Analysis

After the data is cleaned enough, the next step is to do some exploratory analysis by investigating the basic statistic summary or the data distribution. From the schema in Figure 3, it can be seen that there are three main types of data, numeric (integer and double),

string, and Boolean (logical). Thus for each type, we will give different treatments about how to explore the data.

For the numerical data, we decided to just show a summary table containing the basic descriptive statistics, such as mean, median, standard deviation, first quartile, third quartile, minimum, and maximum, about each column. The result can be seen in Figure 4. By comparing the values of the mean and median of each column, we can get a basic understanding of the data distribution. More or less the distribution looks like this:

- The back_legroom, front_legroom, and width columns are heavily skewed to the left.
- The year, wheel_base, fuel_tank_volume, height, and length are skewed to the left.
- daysonmarket, engine_displacement, price, and mileage are skewed to the right.
- city_fuel_economy, highway_fuel_economy, and horsepower have a normal distribution.
- owner_count is a unique case as the minimum, first quartile, median, mean, and third quartile have the same value (around 1), whereas the maximum is much larger. This may happen due to a large number of records having a value of 1, thus heavily skewed the distribution.

summary	back_legroom	city_fuel_economy	daysonmarket	engine_displacement	front_legroom
count	694605	694605	694605	694605	694605
mean	354.2146658892464	21.061470907926086	43.39215525370534	2952.6951288862015	369.37921120636906
stddev	89.56128155575986	4.580394158432613	73.20664937370852	1144.886718194092	132.2751650717123
min	0.0	9	0	1000	0.0
25%	356.0	17	11	2000	408.0
50%	376.0	21	25	2500	416.0
75%	394.0	25	48	3600	428.0
max	598.0	74	2688	8100	525.0

summary	fuel_tank_volume	height	highway_fuel_economy	horsepower	price
count	694605	694605	694605	694605	694605
mean	125.25643351257189	589.1953354784374	28.518294570295346	247.61337018881235	24024.255515911922
stddev	75.1887360603059	188.3376881835074	5.538200724878547	82.07694883453584	12494.583131147947
min	8.0	49.0	11	55	484.0
25%	26.0	573.0	24	178	16109.0
50%	153.0	653.0	28	245	21500.0
75%	185.0	693.0	33	295	29994.0
max	335.0	1086.0	75	808	2698500.0

summary	width	year	mileage	owner_count	length	wheelbase
count	694605	694605	694605	694605	694605	694605
mean	683.1355230670669	2016.7401847092951	46099.33506381325	1.3231491279216245	1750.6762404532071	1036.8719804781135
stddev	253.4702264330892	2.684279766486923	36204.18334553007	0.6563307735021722	526.3323986198891	303.10857424813247
min	66.0	1987	0	1	152.0	97.0
25%	709.0	2016	22712	1	1793.0	1051.0
50%	744.0	2017	35497	1	1884.0	1093.0
75%	818.0	2019	58898	1	1962.0	1142.0
max	986.0	2021	399496	12	2661.0	1646.0

Figure 4. Summary statistics of the numerical columns in the dataset

Other than just investigating the basic descriptive statistics, we also made a boxplot and histogram of the 'price' column to dig deeper into the distribution of the data. The column 'price' is chosen because it will be used as the target variable for our machine learning model, thus it is better to explore more about the column. The boxplot and histogram can be seen in Figure 5 and Figure 6. From the boxplot, it can be seen that the 'price' column has a large number of outliers, especially the ones that are greater than the upper boundary. This causes the distribution of the 'price' in the histogram to look heavily skewed to the right, even though the mean and median should not be that significantly different.

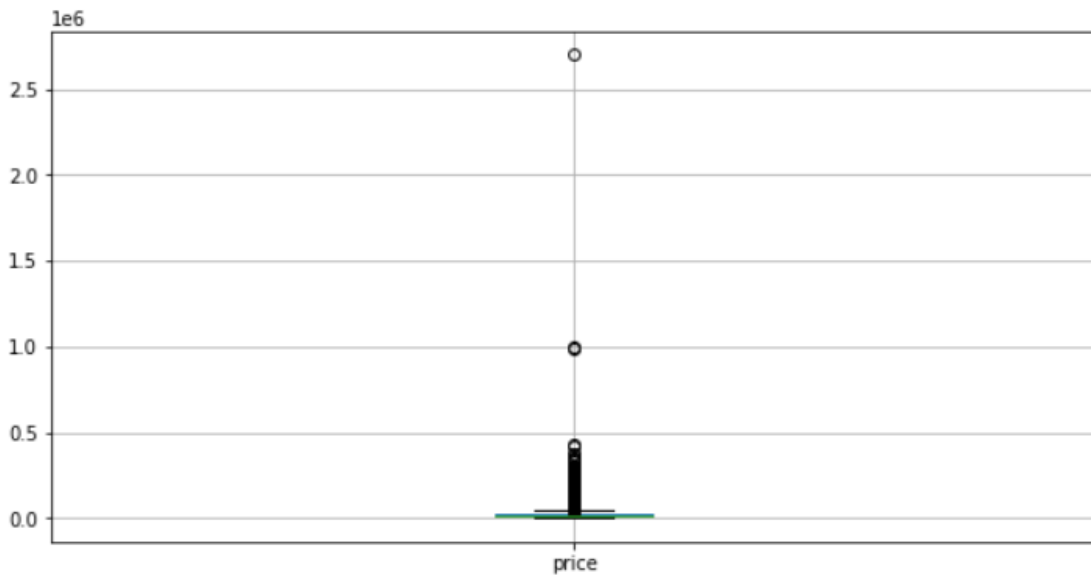


Figure 5. Boxplot of the 'price' column



Figure 6. Histogram of the 'price' column

Then, for the string data type, we only count the frequency (number of occurrences) of each unique value (category) in the column. The result can be seen in Figure 7.

body_type	count	franchise_make	count	fuel_type	count	listing_color	count
SUV / Crossover	347161	Chevrolet	97777	Gasoline	654346	WHITE	142196
Sedan	197005	Ford	94944	Flex Fuel Vehicle	35978	BLACK	140293
Pickup Truck	74484	Jeep	63578	Diesel	3175	SILVER	97855
Minivan	24991	Toyota	57077	Biodiesel	686	GRAY	90852
Coupe	16797	Honda	53988	Hybrid	413	UNKNOWN	87690
Hatchback	14184	Nissan	40724	Compressed Natura...	7	RED	61732
Wagon	14154	Kia	26629			BLUE	58146
Convertible	5072	Hyundai	24885			BROWN	4800
Van	757	Buick	23940			GREEN	4534
		Subaru	21002			ORANGE	2310
		Volkswagen	18797			GOLD	1779
		BMW	18765			TEAL	1070
		Mercedes-Benz	15134			YELLOW	868
		Dodge	14472			PURPLE	444
		Mazda	14073			PINK	36
		GMC	11360				
		Lexus	10941				
		Cadillac	10163				
		Acura	9595				
		Audi	8577				

only showing top 20 rows

power	count	torque	count	transmission	count	wheel_system	count
355 hp @ 5,600 RPM	22889	383 lb-ft @ 4,100...	22889	A	570802	FWD	311793
170 hp @ 6,000 RPM	16580	260 lb-ft @ 4,800...	17308	CVT	108126	AWD	191167
185 hp @ 6,000 RPM	15180	175 lb-ft @ 4,400...	16223	M	13242	4WD	123943
245 hp @ 5,500 RPM	14655	178 lb-ft @ 4,000...	15140	Dual Clutch	2435	RWD	43546
395 hp @ 5,600 RPM	13207	260 lb-ft @ 4,400...	15106			4X2	24156
283 hp @ 6,400 RPM	11045	275 lb-ft @ 3,000...	13851				
132 hp @ 6,000 RPM	10792	410 lb-ft @ 3,950...	12441				
147 hp @ 6,200 RPM	10443	128 lb-ft @ 4,400...	10663				
176 hp @ 6,000 RPM	10137	132 lb-ft @ 4,500...	10613				
395 hp @ 5,750 RPM	9708	172 lb-ft @ 4,100...	9816				
240 hp @ 5,500 RPM	9647	175 lb-ft @ 3,900...	9290				
138 hp @ 4,900 RPM	9492	390 lb-ft @ 4,250...	9071				
180 hp @ 6,400 RPM	9290	255 lb-ft @ 4,000...	8973				
290 hp @ 6,500 RPM	8973	263 lb-ft @ 4,700...	8327				
295 hp @ 6,400 RPM	8964	148 lb-ft @ 200 RPM	8155				
360 hp @ 5,150 RPM	8857	184 lb-ft @ 2,500...	8126				
375 hp @ 5,000 RPM	8447	239 lb-ft @ 4,400...	7369				
181 hp @ 6,000 RPM	8123	138 lb-ft @ 4,200...	7282				
160 hp @ 5,700 RPM	8059	262 lb-ft @ 4,700...	7219				
271 hp @ 6,500 RPM	7369	174 lb-ft @ 4,000...	7165				

only showing top 20 rows only showing top 20 rows

Figure 7. Frequency of each unique value (category) of the string column in the dataset

Lastly, for the Boolean columns, we make a pie chart to compare the percentage of True and False values of each column. The collection of the pie charts can be seen in Figure 8. From those pie charts, the False records dominate every column. This may happen because here we set the False value as the default value to be imputed to the records with missing value. Thus this may skew the actual data distribution.

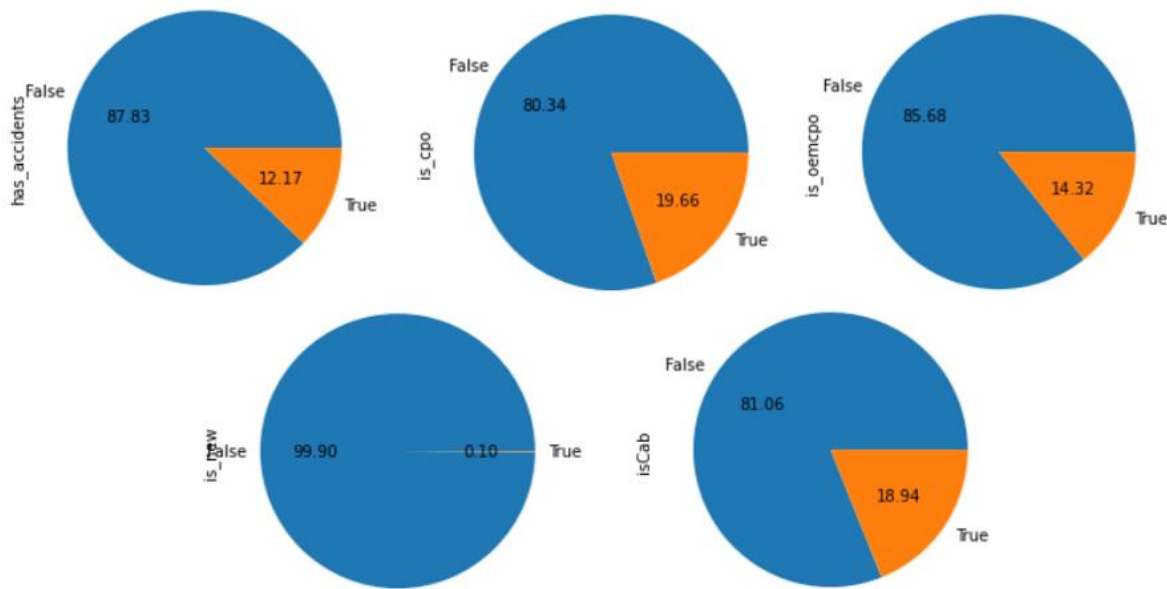


Figure 8. A collection of the pie chart to show the distribution of the Boolean columns

3.4. Machine Learning Plan

The main machine learning process that we are going to do is multiple linear regression. Here, we will create a model to predict the price of a used car based on several predictors that correspond with its condition or features. For the initial stage, we use every column, except 'price', in the subset as the predictors to create the initial model. Because we will use a lot of predictors, we will also use the “k-fold cross-validation” method to minimize the probability of overfitting. The “k-fold cross-validation” will divide the dataset into k partitions and use k-1 segments as the training data and the remaining segment as the test data. Then, that process will be repeated by shuffling which segments are for training and which one is for testing until all possible combination have been used. The final coefficients are then calculated by averaging the resulting coefficients from each iteration.

After we manage to create a reliable model, from the resulting model, we also will do some feature analysis by investigating the variable importance of each predictor. From this feature analysis, we hope to be able to conclude which predictors have a big impact on price-making.

4. Work Division

Part	Eddy	Fariz	Robby
1. Project Description	40%	30%	30%
2. Business Model	30%	30%	40%
3. Data Analysis			
a. Data Wrangling	40%	30%	30%
b. EDA	30%	30%	40%
c. Machine Learning Plan	25%	50%	25%

Note:

The resources used for this project can be accessed using this link:

https://drive.google.com/drive/folders/1Jlx-MaE2cHIc8_LHVJs0CiWAGL22cFw?usp=sharing

References

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